

A Taxonomy of Hallucination Types in Large Language Models for Research Applications

Abstract

Large Language Models (LLMs) are increasingly used across research workflows for literature discovery, synthesis, question answering, coding, data interpretation, summarization, drafting, and citation generation. Their usefulness is constrained, however, by hallucinations: outputs that are fluent and plausible but false, unsupported, internally inconsistent, or unfaithful to an information source or user-provided context. Existing literature has established several related distinctions, including intrinsic versus extrinsic hallucination, factuality versus faithfulness hallucination, and errors arising from model knowledge versus failures of source grounding. Yet these classifications do not fully capture the heterogeneous failure modes encountered in research applications, where fabricated citations, incorrect quantitative claims, temporal knowledge errors, unsupported synthesis, methodological misrepresentation, and erroneous tool-mediated evidence can have consequences beyond ordinary question-answering errors. This paper develops a research-oriented taxonomy that organizes hallucinations along three complementary dimensions: **epistemic relation**, **failure locus**, and **research manifestation**. The resulting framework distinguishes factual hallucinations, source-faithfulness hallucinations, reasoning and inferential hallucinations, citation and bibliographic hallucinations, methodological hallucinations, temporal and version hallucinations, and tool- or retrieval-mediated hallucinations. The paper reviews established detection and mitigation approaches, including factuality metrics, retrieval-augmented generation, self-verification, semantic uncertainty, and citation-aware generation. It argues that no single evaluation paradigm is adequate because hallucination detection is task-, domain-, evidence-, and granularity-dependent. Finally, the paper identifies research gaps concerning benchmark realism, uncertainty calibration, provenance, longitudinal knowledge change, domain-specific evaluation, causal diagnosis, and human–LLM verification. A research-oriented taxonomy can therefore serve not merely as a descriptive classification but as a basis for choosing evaluation protocols, designing safeguards, and specifying trustworthy uses of LLMs in scholarly research.

Keywords: large language models; hallucination; factuality; faithfulness; research integrity; citation hallucination; retrieval-augmented generation; uncertainty estimation; provenance; scholarly AI

1. Introduction

Large Language Models have transformed natural-language processing from a collection of narrowly specified prediction tasks into general-purpose systems capable of answering questions, summarizing documents, generating explanations, writing code, and interacting with external tools. These capabilities have rapidly expanded their use in scholarly and professional research workflows. Researchers increasingly employ LLMs for literature exploration, summarization, drafting, data extraction, methodological assistance, hypothesis generation, and interaction with scientific databases. At the same time, the models remain probabilistic generators rather than databases of verified propositions. Consequently, fluent language can coexist with inaccurate or unsupported claims.

The literature commonly uses the term *hallucination* for generated content that is factually incorrect, unverifiable, or inconsistent with a source or instruction (Ji et al., 2023; Huang et al., 2024). Earlier work on neural text generation emphasized hallucination in task-specific settings such as summarization and data-to-text generation, where the central issue was whether generated text was supported by an input source (Maynez et al., 2020). With the emergence of open-ended LLMs, the problem has broadened. An LLM may produce a false statement even when no explicit source is provided, contradict an authoritative source supplied in the prompt, fabricate a publication that appears bibliographically plausible, misattribute a finding to an authentic article, or combine individually plausible observations into an unsupported scientific conclusion.

This diversity presents a taxonomy problem. A classification designed for abstractive summarization does not adequately distinguish a fabricated DOI from an incorrect interpretation of an authentic paper. Conversely, a simple factual/non-factual distinction is insufficient for applications in which the relevant question is not only whether a statement is true, but whether it follows from a cited source, whether the model has accurately represented a methodology, or whether the evidence was available at the claimed date.

Two broad conceptual distinctions are especially influential. The first is **factuality**, concerning agreement between generated claims and external reality or verifiable evidence. The second is **faithfulness**, concerning whether an output appropriately follows a supplied source, context, or instruction (Huang et al., 2024). Earlier summarization research similarly distinguished *intrinsic* errors, which contradict information in a source, from *extrinsic* errors, which introduce information not supported by that source (Maynez et al., 2020). These distinctions remain useful, but research applications require additional categories because the scholarly object being represented may itself be a paper, citation, dataset, experiment, result, method, or historical state of knowledge.

This paper therefore proposes a taxonomy specifically oriented toward research applications. Its central argument is that hallucination should be represented as a **multidimensional phenomenon rather than a single error class**. In particular, the paper distinguishes: (1) the relation between a claim and the relevant truth or evidence, (2) the stage or mechanism at which the failure arises, and (3) the concrete research artifact or activity affected by the failure.

This perspective is important because mitigation strategies are not interchangeable. Retrieval can reduce unsupported factual claims but cannot guarantee correct interpretation of a retrieved paper. Self-consistency can identify some uncertainty but cannot establish bibliographic existence. Citation verification can detect fabricated references but does not ensure that an authentic citation actually supports the associated statement. A robust research assistant therefore requires a layered architecture of retrieval, provenance, verification, uncertainty estimation, and human oversight.

2. Background and Related Work

2.1 From neural-generation hallucination to LLM hallucination

Hallucination predates contemporary LLMs. In neural machine translation and abstractive summarization, generated text could be fluent while introducing entities, facts, or relationships absent from the source. Maynez et al. (2020) demonstrated that abstractive summarization systems can produce substantial amounts of unfaithful content and that standard lexical metrics such as ROUGE are insufficient for reliably measuring factual faithfulness. This work helped establish textual entailment and source-grounded evaluation as alternatives to surface-form overlap.

Kryscinski et al. (2020) introduced FactCC, a factual-consistency model trained to identify whether summary sentences are supported by their source documents. Goyal and Durrett (2020) further proposed dependency-level entailment to localize factual inconsistencies rather than reducing an entire output to a sentence-level judgment. Other approaches employed question generation and question answering to compare what a source and summary imply about the same entities or claims.

These developments established a foundational insight: **fluency is not evidence of factuality**. A model can produce highly coherent language while failing to preserve the truth conditions of the source.

2.2 Taxonomies of LLM hallucination

Ji et al. (2023) provided a broad survey of hallucination in natural language generation and synthesized distinctions across translation, dialogue, summarization, and other tasks. More recent surveys have adapted the taxonomy to general-purpose LLMs. Huang et al. (2024), for example, distinguish **factuality hallucination** from **faithfulness hallucination** and organize hallucination causes around data, training, and inference factors. Their framework highlights the fact that hallucinations can originate from limitations in training data and memorized knowledge, alignment objectives, prompting, decoding, or contextual grounding.

The distinction is valuable but does not by itself specify what kind of research activity has failed. A statement such as "Smith et al. (2021) reported a significant effect" can be wrong because the paper does not exist, because the paper exists but did not report the effect, because the result was associated with a different study, or because the underlying effect was statistically misunderstood. All four cases may be classified as factual or faithfulness failures, but their detection and remediation requirements differ substantially.

2.3 Factuality and truthfulness

Truthfulness benchmarks such as TruthfulQA examine whether language models repeat common falsehoods and misconceptions learned from human text (Lin et al., 2022). Such work illustrates an important mechanism of hallucination: a model can reproduce plausible linguistic patterns associated with false propositions even when the propositions themselves are incorrect. In TruthfulQA, model scale did not automatically guarantee truthfulness; some larger systems performed worse on the benchmark than smaller models.

FactScore introduced a more fine-grained approach for long-form generation by decomposing text into atomic facts and estimating what proportion of those facts are supported by a reliable knowledge source (Min et al., 2023). This approach is particularly relevant to research writing because an apparently reliable paragraph can contain a mixture of correct, partially correct, and fabricated claims. Evaluating at the atomic-claim level therefore provides more diagnostic information than assigning one binary factuality label to an entire response.

2.4 Detecting hallucinations through uncertainty and consistency

Another research direction asks whether the model itself contains signals indicating when it is likely to be wrong. Kadavath et al. (2022) studied whether language models can assess the validity of their own claims and introduced probability-based measures such as "P(True)" and "P(IK)" ("I know"). Their findings suggested that models can possess useful but imperfect information about their own uncertainty.

Farquhar et al. (2024) developed **semantic entropy**, which estimates uncertainty at the level of meaning rather than surface wording. Their results showed that semantic diversity among sampled answers can provide a signal for hallucination, particularly for questions where the model lacks reliable knowledge. Such approaches shift hallucination detection from post hoc fact checking toward uncertainty-aware generation and selective prediction.

Manakul et al. (2023) introduced SelfCheckGPT, a black-box method that samples multiple responses and identifies likely hallucinations when supposedly factual claims vary substantially across samples. The method illustrates another important principle: **inconsistency across semantically equivalent generations can be evidence of epistemic uncertainty**, although consistency alone cannot establish truth.

3. A Research-Oriented Taxonomy of LLM Hallucinations

3.1 Taxonomy design principles

For research applications, a useful taxonomy should satisfy four criteria. First, categories should be **operationally distinguishable**, so that different classes can be evaluated with different methods. Second, the taxonomy should accommodate both **open-domain factual claims and source-grounded research tasks**. Third, it should distinguish errors in *content* from errors in *evidence representation*. Fourth, it should support **mitigation mapping**, allowing researchers to identify which controls can address a particular failure.

The proposed framework uses three complementary dimensions:

1. **Epistemic relation:** What is the relationship between the output and the truth/evidence?
2. **Failure locus:** At what point in the generation pipeline does the error arise?
3. **Research manifestation:** What research artifact or activity does the error affect?

The categories below are therefore not necessarily mutually exclusive. A single output may simultaneously contain a source-faithfulness hallucination and a citation hallucination.

3.2 Type I: Factual or world-knowledge hallucination

A **factual hallucination** is a claim that conflicts with verifiable external knowledge or lacks adequate factual support when presented as fact.

Examples include incorrect dates, invented people, wrong numerical values, false historical statements, or misidentified scientific findings. This is the most familiar form of hallucination and corresponds closely to factuality-oriented definitions in recent surveys (Ji et al., 2023; Huang et al., 2024).

Within research contexts, factual hallucinations are especially consequential when they concern:

- numerical results;
- sample sizes;
- experimental outcomes;
- statistical significance;
- dataset properties;
- author names;
- publication dates;
- institutional affiliations;
- scientific definitions; or
- technical specifications.

Factual errors may be **atomic**, affecting one proposition, or **compositional**, where several small inaccuracies accumulate into a misleading explanation. FActScore's atomic decomposition is particularly relevant because research prose often contains many independently checkable propositions (Min et al., 2023).

3.3 Type II: Source-faithfulness hallucination

A **source-faithfulness hallucination** occurs when generated content is inconsistent with, unsupported by, or improperly derived from an identified source.

This category is central to literature reviews, document summarization, systematic reviews, and evidence synthesis. A source may state that an intervention was *associated with* an outcome while the LLM summarizes it as demonstrating *causation*. Similarly, a paper may report a result only for one subgroup while the model generalizes it to an entire population.

The distinction from ordinary factual hallucination is important. A generated statement can even be factually true in the real world while being **unfaithful to the cited source**. For example, a claim may be scientifically established elsewhere but not actually reported by the article being summarized.

The intrinsic/extrinsic distinction from summarization research remains useful here. Intrinsic errors contradict source information, whereas extrinsic errors introduce unsupported information (Maynez et al., 2020).

3.4 Type III: Citation and bibliographic hallucination

A **citation hallucination** involves fabricated, malformed, or improperly attributed scholarly references. It includes at least four subtypes:

1. **Existence hallucination:** the cited paper, book, dataset, or DOI does not exist.
2. **Metadata hallucination:** the work exists, but authors, title, venue, date, pages, or identifier are incorrect.
3. **Attribution hallucination:** a real work is cited for a claim it does not support.
4. **Citation-context hallucination:** an authentic reference is inserted into a paragraph in a way that falsely suggests evidentiary support.

This category deserves explicit recognition in research-oriented taxonomies because bibliographic references are not merely prose decorations; they are part of the evidentiary infrastructure of scholarship.

Empirical studies demonstrate the seriousness of the problem. Walters and Wilder (2023) found that, across their evaluated literature-review prompts, fabricated references were common, although GPT-4 substantially reduced fabrication relative to GPT-3.5. Gravel et al. (2023) similarly reported a high proportion of fabricated references in answers to medical questions.

These findings demonstrate that citation hallucination can remain present even when the generated prose appears professionally written.

A useful extension is **identifier hallucination**, in which a model supplies a plausible but nonexistent DOI, PMID, ISBN, arXiv identifier, or URL. Identifier verification is therefore an important component of scholarly LLM evaluation.

3.5 Type IV: Reasoning and inferential hallucination

A **reasoning hallucination** occurs when the model derives an unsupported conclusion from otherwise plausible premises. Unlike a simple factual error, the individual statements may be accurate while the inference connecting them is invalid.

Examples include:

- confusing correlation with causation;
- inferring statistical significance from a large effect size without considering uncertainty;
- treating absence of evidence as evidence of absence;
- inferring generalizability from a narrow sample;
- converting an observational finding into an experimental conclusion;
- claiming that two studies are contradictory when they use different populations or measures.

This category is especially important for research assistance because LLMs are frequently asked not only to summarize findings but to compare studies, identify causes, assess evidence quality, and generate hypotheses.

Reasoning hallucinations challenge simple fact-checking because verifying individual propositions may not reveal the inferential failure. Evaluation must therefore sometimes operate on **argument structure**, assumptions, and logical entailment.

3.6 Type V: Methodological hallucination

A **methodological hallucination** occurs when an LLM incorrectly represents a research method, experimental procedure, dataset construction process, statistical technique, or analytical protocol.

Examples include attributing a randomized controlled trial design to an observational study, stating that a paper used logistic regression when it used a survival model, or claiming that a dataset contains variables that are absent from its published documentation.

Methodological hallucination deserves a separate class because methods are often used as the basis for judging evidence quality. An incorrect representation can therefore distort not only one fact but the researcher's entire assessment of a study.

A related problem is **procedural invention**, where the model fills missing experimental details with plausible-sounding assumptions. Such details may be particularly difficult to identify because the invented method can be scientifically reasonable even though it was never performed.

3.7 Type VI: Temporal, version, and knowledge-boundary hallucination

A **temporal hallucination** arises when a model incorrectly represents the state of knowledge at a particular time. Examples include citing a result as available before its publication, treating a draft or preprint as a peer-reviewed article, applying a current standard to a historical study, or describing a software library's current behavior as though it existed in an earlier version.

This category follows from the fact that research knowledge is dynamic. A statement can be true in 2026 and false as a description of the scientific record in 2021.

Temporal hallucination should therefore be distinguished from ordinary factual error. The central question is not simply "Is this statement true?" but **"Was this statement true, established, or knowable within the specified temporal frame?"**

3.8 Type VII: Retrieval- and tool-mediated hallucination

Retrieval-Augmented Generation (RAG) can reduce reliance on parametric knowledge by supplying external evidence (Lewis et al., 2020), but retrieval does not eliminate hallucination. Instead, it introduces new error pathways:

- retrieving an irrelevant document;
- retrieving outdated information;
- retrieving a secondary source when a primary source is required;
- misreading retrieved evidence;
- ignoring contradictory passages;
- attributing information to the wrong document; or
- generating claims not supported by retrieved text.

Thus a model can hallucinate **because retrieval failed or despite successful retrieval**.

RAGAS explicitly treats retrieval relevance, faithfulness to retrieved context, and generation quality as distinct evaluation dimensions (Es et al., 2024).

Corrective Retrieval-Augmented Generation (CRAG) and Self-RAG illustrate attempts to address these weaknesses. CRAG evaluates retrieved material and triggers corrective retrieval when necessary, while Self-RAG trains the model to retrieve, generate, and critique selectively (Yan et al., 2024; Asai et al., 2024).

3.9 Type VIII: Self-verification and tool-use hallucination

A newer class concerns hallucinations about the model's **own actions**. An LLM may claim that it searched a database, opened a paper, executed code, checked a DOI, or verified a result when it did not actually perform that operation.

This is conceptually distinct from ordinary factual hallucination because the false proposition concerns **process provenance**. Research environments increasingly involve tool-augmented agents, making this category increasingly relevant.

A trustworthy research system must therefore distinguish between:

- evidence actually retrieved;
- evidence supplied by the user;
- information inferred from model parameters; and
- verification steps that were actually executed.

A model should not be allowed to convert an intended action into a claimed completed action.

3.10 Type IX: Synthesis and attribution hallucination

Research commonly involves integrating multiple papers rather than summarizing one source. A **synthesis hallucination** occurs when the model constructs a cross-paper conclusion that is not justified by the collection of sources.

For example, a model may correctly report results from Studies A, B, and C but then assert that "the literature consistently demonstrates X" even though the studies use incompatible definitions or reach divergent conclusions.

This category is distinct from simple citation misattribution because all cited papers may be genuine and correctly represented individually. The hallucination occurs at the **level of synthesis**.

4. Causes and Mechanisms

Hallucinations arise from interacting properties of data, learning objectives, model architecture, inference, and task design (Ji et al., 2023; Huang et al., 2024).

4.1 Imperfect and heterogeneous training data

Web-scale corpora contain misinformation, contradictions, duplicated material, outdated information, low-quality sources, and incomplete documentation. A language model trained to predict text is therefore not directly optimized to distinguish reliable knowledge from unreliable text.

TruthfulQA demonstrates one consequence: models may reproduce common misconceptions because those misconceptions are represented in their training distribution (Lin et al., 2022).

4.2 Parametric knowledge and incomplete retrieval

LLMs encode information indirectly in parameters. This knowledge can be useful but may be incomplete, outdated, entangled across contexts, or difficult to access precisely.

Knowledge-intensive tasks motivated the development of RAG precisely because parametric models have limitations in accessing and updating factual information (Lewis et al., 2020).

4.3 Objective mismatch

The training objective for a conventional autoregressive model rewards likely continuations rather than truth. Fluent continuation therefore does not necessarily imply factual verification.

This mismatch also explains why increasing model size alone cannot be assumed to solve hallucination. Larger models may improve many capabilities while retaining systematic weaknesses in truthfulness and calibration.

4.4 Instruction following and alignment pressure

Alignment methods encourage models to satisfy user requests and produce helpful responses. In some circumstances, however, a strong pressure to answer can compete with the safer behavior of explicitly stating uncertainty or refusing to speculate.

Research prompts frequently contain implicit expectations such as "provide the references," "explain why," or "identify the studies." If the model does not possess sufficient information, the objective of being helpful can produce plausible completion rather than abstention.

4.5 Decoding and stochasticity

Sampling-based generation permits multiple plausible continuations. SelfCheckGPT exploits this property by treating substantial disagreement among sampled outputs as a possible signal of hallucination (Manakul et al., 2023). More generally, generation settings can influence factual consistency, although decoding changes alone cannot solve underlying knowledge or grounding problems.

5. Current Approaches to Detection and Mitigation

5.1 Human verification

Human expert review remains the strongest general-purpose safeguard for high-stakes research claims, but it is expensive and difficult to scale. Human reviewers also exhibit domain and source-selection biases. A practical system therefore needs automated screening to prioritize human attention.

5.2 Claim-level factuality evaluation

FactScore represents an important shift toward atomic claim evaluation. Rather than asking whether an entire response is factual, the evaluation estimates the fraction of atomic statements that are independently supported (Min et al., 2023).

This approach is particularly appropriate for research assistants because a single paragraph can contain multiple claims about authors, methods, datasets, results, and interpretations.

5.3 Entailment- and QA-based evaluation

FactCC, dependency-level entailment, and question-answering approaches evaluate whether a generated statement follows from a reference document (Kryscinski et al., 2020; Goyal & Durrett, 2020). These methods are useful for summarization and source-grounded generation, but their performance can degrade when claims are complex, implicit, multi-document, or highly specialized.

5.4 Sampling-based self-consistency

SelfCheckGPT uses multiple generations to identify information that is unstable across samples (Manakul et al., 2023). This is attractive for black-box systems because it does not require access to model probabilities or an external knowledge base.

However, self-consistency should not be confused with correctness. A model may repeatedly produce the same false statement because that statement is strongly encoded in its parameters.

5.5 Semantic uncertainty

Semantic entropy is designed to aggregate uncertainty over meanings rather than token sequences. Farquhar et al. (2024) showed that semantic entropy can detect an important class of hallucinations associated with lack of model knowledge.

Its research value lies in supporting **selective answering**. Instead of requiring the system to answer every question, the model can abstain or trigger external verification when semantic uncertainty is high.

5.6 Retrieval-Augmented Generation

RAG connects a generator to an external corpus, allowing the system to use information beyond its parametric memory (Lewis et al., 2020). For research applications, this can facilitate access to current literature and provide source provenance.

Yet RAG must not be treated as synonymous with factuality. Retrieval introduces errors of relevance, coverage, freshness, ranking, and interpretation. A retrieved passage can itself be low quality or contradictory.

RAGAS explicitly decomposes evaluation into retrieval quality, context utilization/faithfulness, and generation quality (Es et al., 2024). Self-RAG further allows a model to decide when retrieval is needed and to critique its own evidence and generation (Asai et al., 2024). CRAG introduces a retrieval-quality evaluator and corrective retrieval procedures (Yan et al., 2024).

5.7 Chain-of-Verification and post-generation verification

Chain-of-Verification (CoVe) first drafts a response, generates verification questions, answers those questions independently, and then constructs a revised answer (Dhuliawala et al., 2023). This architecture addresses an important failure mode: a single generation may reinforce its own unsupported claims.

Verification, however, remains vulnerable when the model is both generator and verifier. Independent retrieval, external databases, symbolic tools, or human review can provide stronger epistemic independence.

5.8 Citation verification

Research-oriented LLMs should treat references as structured data requiring verification rather than as unconstrained text generation. A reliable pipeline should validate, when possible:

- publication existence;
- author–title consistency;
- venue and year;
- DOI or other identifier;
- bibliographic metadata; and
- whether the source actually supports the cited claim.

Evidence from fabricated-citation studies demonstrates why this layer is necessary (Walters & Wilder, 2023; Gravel et al., 2023).

6. Comparative View of Hallucination Types

Hallucination type	Primary question	Typical research example	Best-suited detection
Factual	Is the claim true?	Incorrect statistic	External fact checking, claim-level verification
Source-faithfulness	Is the claim supported by the cited/source text?	Misrepresenting a paper's result	Entailment, QA, source comparison
Citation/bibliographic	Does the cited source exist and contain the claimed metadata?	Fabricated DOI	Crossref/Scopus/PubMed/arXiv-style verification
Reasoning/inferential	Does the conclusion follow from the premises?	Correlation → causation	Argument analysis, expert review
Methodological	Is the study method represented correctly?	Calling an observational study randomized	Full-text/source inspection
Temporal/version	Is the claim correct for the specified date/version?	Attributing later knowledge to an earlier paper	Time-aware retrieval and version tracking
Retrieval/tool-mediated	Did external evidence or a tool actually support the answer?	Irrelevant retrieved paper	Retrieval diagnostics, provenance
Self-verification/tool-use	Did the system actually perform the claimed verification?	"I checked the DOI" when no check occurred	Execution logs and tool provenance

Synthesis/attributions	Does the multi-source conclusion accurately integrate the literature?	Claiming consensus from conflicting studies	Cross-document evidence graphs
------------------------	---	---	--------------------------------

The table illustrates why the binary label "hallucination/no hallucination" is insufficient for research deployment. Different categories require different evidence and different controls.

7. Research Challenges and Limitations

7.1 No universally accepted operational definition

Hallucination remains a broad term covering several conceptually different phenomena. A model output may be false, unverifiable, unsupported by a provided source, logically invalid, or merely uncertain. Treating all such cases as equivalent complicates benchmarking and makes results difficult to compare across studies.

A robust evaluation protocol should therefore declare the **reference standard**: world knowledge, source document, database record, scientific consensus, or a specified temporal snapshot.

7.2 Evaluation depends on granularity

Whole-response evaluation can hide substantial local errors. Conversely, atomic-claim evaluation may fragment complex methodological or inferential statements into pieces that appear individually correct despite a flawed overall argument.

Future benchmarks should therefore operate at multiple granularities:

1. token/entity level;
2. atomic claim level;
3. sentence level;
4. argument level;
5. source/document level; and
6. research-conclusion level.

7.3 Automatic evaluators can hallucinate

An important limitation is evaluator reliability. LLM-based judges may themselves produce incorrect judgments, creating an evaluator-generator dependency. Even entailment models can

fail when claims involve specialized terminology, numerical reasoning, negation, or implicit relationships.

Accordingly, evaluation systems should report agreement with expert human judgments and test the robustness of automatic metrics rather than treating model-generated scores as ground truth.

7.4 Domain expertise remains indispensable

Scientific and technical hallucinations often involve subtle distinctions that general-purpose evaluators cannot reliably recognize. Whether an inference is justified may depend on domain conventions, statistical assumptions, experimental design, or specialized vocabulary.

A generic hallucination detector may therefore perform well on Wikipedia-style questions while failing on specialized biomedical, legal, engineering, or social-science content.

7.5 Bibliographic verification is not enough

A reference can be authentic while the associated claim is unsupported. Therefore, bibliographic validation should be viewed as only one stage in a citation-verification pipeline.

Research systems should distinguish at least:

source existence → metadata correctness → source accessibility → claim support → interpretation correctness.

7.6 RAG creates a false sense of security

RAG improves access to external information but can introduce retrieval-specific failure modes. Poor retrieval, incomplete evidence, context overload, or incorrect interpretation can still produce hallucinated conclusions. RAG systems should therefore expose retrieval provenance and evidence coverage rather than merely displaying a list of sources.

7.7 Temporal drift

Research information changes over time. Benchmarks that use static truth sets may become outdated, especially for software, medical knowledge, policies, standards, and rapidly developing scientific areas.

Future datasets should support versioned or date-indexed ground truth.

8. Research Gaps and Future Directions

8.1 Standardized multidimensional benchmarks

Current benchmarks frequently focus on one dimension—truthfulness, factual consistency, or hallucination recognition. Future research should build **multidimensional research hallucination benchmarks** in which each example is annotated for:

- factual correctness;
- source support;
- citation correctness;
- methodological accuracy;
- inferential validity;
- temporal validity;
- evidence provenance; and
- uncertainty/abstention appropriateness.

This would make it possible to compare systems across genuinely different failure modes.

8.2 Provenance-native language models

Research assistants should not merely produce answers; they should maintain explicit provenance graphs connecting each claim to its evidence. A future response representation could encode:

```
[  
  \text{Claim} \rightarrow \text{Evidence Passage} \rightarrow \text{Document} \rightarrow  
  \text{Identifier} \rightarrow \text{Verification State}.  
]
```

Such representations could turn hallucination prevention from a purely linguistic problem into an information-provenance problem.

8.3 Claim-level uncertainty and calibrated abstention

Future systems should estimate confidence at the level of individual claims rather than entire responses. A response might contain ten claims where eight are highly supported and two require verification.

This suggests an architecture in which the model can produce:

- supported;
- weakly supported;
- conflicting evidence;
- unverifiable; or
- high-uncertainty

labels alongside claims.

Semantic entropy and self-evaluation research provide promising foundations, but stronger calibration and external validation are needed (Kadavath et al., 2022; Farquhar et al., 2024).

8.4 Temporal and version-aware research assistants

Scholarly systems should explicitly model time. Literature agents should distinguish a preprint from its peer-reviewed version, identify retractions and corrections, and preserve the state of evidence at the date relevant to the research question.

Temporal provenance is particularly important for systematic reviews and longitudinal scientific analyses.

8.5 Citation verification as an integrated architecture

Citation checking should occur during generation rather than only after a paper has been drafted. Future systems could retrieve candidate sources, verify bibliographic records, identify supporting passages, and prohibit unsupported citations from appearing in the final manuscript.

The empirical evidence on fabricated citations strongly motivates this direction (Walters & Wilder, 2023). More recent scholarship on reference hallucinations further suggests that citation constraints and task structure can materially influence the form of hallucination produced by LLMs (Davis & Mahmoud, 2026).

8.6 Independent verification agents

A generator should not necessarily be its own final verifier. Future architectures could employ specialized independent agents:

- a **retrieval agent** for evidence discovery;
- a **bibliographic agent** for metadata verification;
- a **factuality agent** for claim checking;
- a **reasoning agent** for inferential validity;
- a **methodology agent** for experimental-design interpretation; and
- a **provenance controller** to ensure that claims are traceable.

Such modular verification could be more reliable than asking a single general-purpose model to "check its work."

8.7 Researcher-centered human–AI interaction

The goal should not necessarily be zero hallucination, which may be unrealistic for open-ended generation. A more practical target is **risk-aware assistance**. Systems should surface uncertain

claims, contradictory evidence, missing citations, and unverifiable statements instead of silently filling gaps.

In scholarly settings, an LLM that occasionally abstains transparently may be more useful than one that generates polished but unverified prose.

8.8 Evaluating downstream research harm

Future work should move beyond measuring hallucination rates and investigate consequences. A fabricated citation in a casual brainstorming session may be inconvenient; the same error in a systematic review, clinical guideline, or meta-analysis can contaminate downstream evidence.

Benchmarks should therefore evaluate not only **frequency**, but **severity, detectability, propagation, and downstream impact**.

9. Conclusion

Hallucination in Large Language Models is not a single failure mode. For research applications, it is a family of failures involving truth, evidence, inference, provenance, methodology, time, retrieval, and scholarly attribution. Existing distinctions such as intrinsic versus extrinsic hallucination and factuality versus faithfulness provide essential foundations, but they do not fully capture the complexity of research-oriented use.

This paper proposed a multidimensional taxonomy comprising factual, source-faithfulness, citation/bibliographic, reasoning, methodological, temporal/version, retrieval- and tool-mediated, self-verification, and synthesis hallucinations. The taxonomy emphasizes that different hallucination types require different forms of evidence and different mitigation strategies.

Current approaches—including FActScore, SelfCheckGPT, semantic entropy, RAG, Self-RAG, CRAG, and Chain-of-Verification—demonstrate substantial progress. Nevertheless, their effectiveness is conditional on task formulation, reference quality, domain expertise, and evaluator reliability. No existing method guarantees that an LLM-generated research output is simultaneously factually correct, source-faithful, methodologically accurate, logically valid, temporally appropriate, and bibliographically authentic.

The central implication for research applications is therefore architectural rather than merely model-centric: trustworthy LLM assistance requires **claim-level evaluation, explicit provenance, retrieval verification, citation checking, uncertainty estimation, temporal awareness, and appropriately targeted human oversight**. Future research should treat hallucination taxonomy not simply as a catalog of mistakes but as a design framework linking identifiable failure modes to evidence standards, evaluation procedures, and mitigation mechanisms. Such a framework is necessary for moving LLMs from fluent research assistants toward dependable components of scholarly knowledge production.

